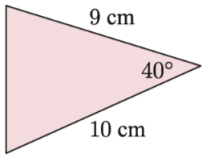
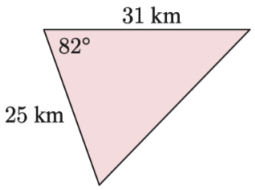
EXERCISE 1 Find the area of :

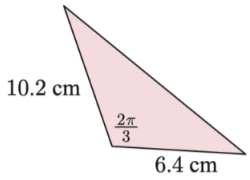
a



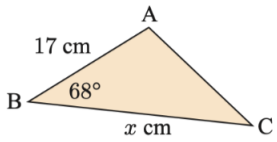
b



c



EXERCISE 2 If triangle ABC has area 150 cm^2, find the value of x.



EXERCISE 3 A parallelogram has two adjacent sides with lengths 4 cm and 6 cm respectively. If the included angle measures $52°$, find the area of the parallelogram.

EXERCISE 4 A rhombus has sides of length 12 cm and an angle of $72°$. Find its area.

EXERCISE 5 Find the area of a regular hexagon with sides of length 12 cm.

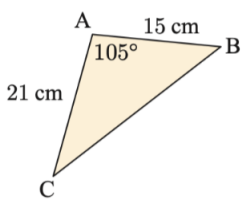
EXERCISE 6 A rhombus has an area of 50 cm^2 and an internal angle of size $63°$. Find the length of its sides.

EXERCISE 7 The Australian 50 cent coin has the shape of a regular dodecagon, which is a polygon with 12 sides. Eight of these 50 cent coins will fit exactly on an Australian $10 note. What fraction of the $10 note is *not* covered ?

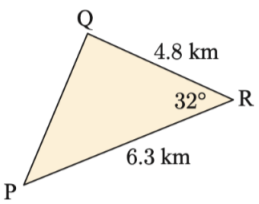


EXERCISE 8 Find the length of the remaining side in the given triangle :

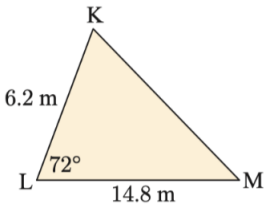
a



b

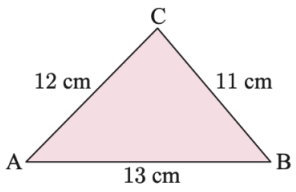


c



EXERCISE 9

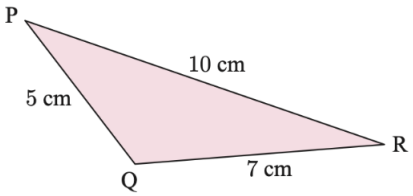
Find the measure of all angles of the triangle :



EXERCISE 10

a. Find the measure of obtuse $\widehat{PQR}$.

b. Hence find the area of $\triangle PQR$.



EXERCISE 11

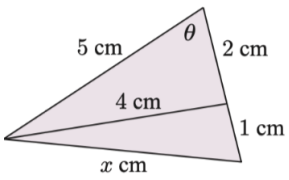
a. Find the smallest angle of a triangle with sides 11 cm, 13 cm, and 17 cm.

b. Find the largest angle of a triangle with sides 4 cm, 7 cm, and 9 cm.

EXERCISE 12

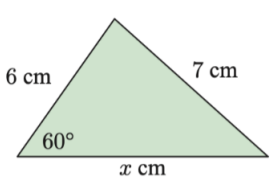
a. Find $\cos\theta$ but not θ.

b. Find the value of x.

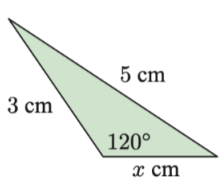


EXERCISE 13 Find the exact value of x in each of the following diagrams :

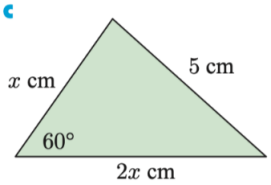
a



b



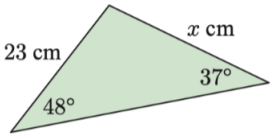
c



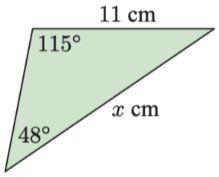
EXERCISE 14

Find the value of x :

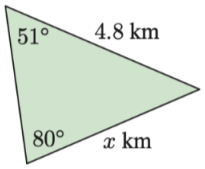
a



b



c



EXERCISE 15 Consider triangle ABC.

a. Given $A = 63°$, $B = 49°$, and $b = 18$ cm, find a.

b. Given $A = 82°$, $C = 25°$, and $c = 34$ cm, find b.

c. Given $B = 21°$, $C = 48°$, and $a = 6.4$ cm, find c.

EXERCISE 16 Triangle ABC has angle $B = 40°$, $b = 8$ cm, and $c = 11$ cm. Find the two possible measures of angle C.

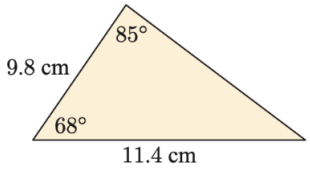
EXERCISE 17 Consider triangle ABC.

a. Given $a = 14.6$ cm, $b = 17.4$ cm, and $\widehat{ABC} = 65°$, find the measure of $\widehat{BAC}$.

b. Given $b = 43.8$ cm, $c = 31.4$ cm, and $\widehat{ACB} = 43°$, find the measure of $\widehat{ABC}$.

c. Given $a = 6.5$ km, $c = 4.8$ km, and $\widehat{BAC} = 71°$, find the measure of $\widehat{ACB}$.

EXERCISE 18 Is it possible to have a triangle with the measurements shown ? Explain your answer.



EXERCISE 19 Find x and y in the given figure.

